Spatially Averaged Turbulent Flow over Square Ribs

S. E. Coleman[1]; V. I. Nikora[2]; S. R. McLean[3]; and E. Schlicke[4]

Abstract: A series of experiments was undertaken to assess fully rough turbulent subcritical flow over two-dimensional transverse repeated-rib roughness of varying spacing $\lambda/h=1-16$ (roughness spacing/height ratio), with $h/H=0.09$ (roughness height/flow depth). Each of the 11 experiments involved centerline measurements of three-dimensional velocity vectors, water surface profiles, and bed pressures and forces. The structure of flow over rib roughness consists of a pair of principal vortices of opposite sign set up by the ribs. These vortices are superimposed on an overall double- (time and space)-averaged velocity profile that is (quasi-) logarithmic above roughness tops, and that below roughness tops changes with increasing rib spacing from exponential ($\lambda/h<10$) to linear ($\lambda/h\geq 10$) to logarithmic ($\lambda/h\gg 10$). The measured double-averaged velocity profiles are parameterized herein, and double-averaged Reynolds and form-induced stress profiles and trends are also identified. Maximum drag due to wall roughness is found to occur for $\lambda/h\approx 8$. The experimentally determined momentum balance, including effects of secondary currents and flow acceleration, is found to agree well with theoretical expectations. It is shown that the knowledge of the effects of form-induced stresses, secondary currents, and flow nonuniformity may be particularly important for describing and modeling flows over roughness elements.

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CE Database subject headings: Turbulent flow; Boundary layers; Bed roughness; Velocity profile; Fluid dynamics; Spatial analysis.

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Introduction

Two-dimensional repeated-rib roughness has many engineering applications, often being used to enhance mass and heat exchange. For turbulent flow over a wall roughened by two-dimensional ribs, Perry et al. (1969) divided wall roughness into two types dependent on the apparent length scale influencing the flow: widely spaced "*k*-type," of length scale roughness height h (k in their work); and closely spaced "*d*-type," of length scale boundary-layer thickness δ (mean flow depth $\bar{H}$ here and pipe diameter d in their work) and not h. Simpson (1973) proposed *d*-type and *k*-type roughnesses to occur for $\lambda/h<5$ and $\lambda/h>5$, respectively, where λ=crest-to-crest rib spacing (Fig. 1). Tani (1987) suggested *d*-type roughness to occur for closely spaced two-dimensional ribs of $\lambda/h<5$, with *k*-type roughness occurring for larger rib spacing. Bandyopadhyay (1987) found $\lambda/h=4$ is small enough to lead to *k*-type roughness; Jiménez (2004) quoted an equivalent value of $\lambda/h=4-5$.

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Perry et al. (1969), Simpson (1973), Stoesser and Rodi (2004), and others, propose that for closely spaced *d*-type roughness (Fig. 1), stable separated vortices occupy the entire cavity between ribs, with negligible eddy shedding from the roughness, and minimal disturbance of the outer flow by the roughness elements and flow within the cavities. The roughness height, hidden from the flow above, thereby does not influence the flow profile. Cui et al. (2000) similarly note flow within the cavity to be akin to two-dimensional lid-driven flow, with velocity, Reynolds stress, and turbulent kinetic energy (TKE) very small within the cavity, and TKE and Reynolds stress reaching maxima a short distance above the rib. In contrast to the above flow patterns, Djenidi et al. (1999) found strong evidence of interactions, albeit intermittent, between cavities and outer flow for their study of turbulent flow over transverse square cavities. Townes and Sabersky (1966) found similar communication for their *d*-type rough wall; these authors, along with Djenidi et al. (1999) and Jiménez (2004), proposed such interactions to potentially arise due to outer-flow variables, these variables then primarily influencing the flow profile in lieu of h. Velocity profiles above the ribs for *d*-type (skimming-flow or quasi-smooth) roughnesses are then essentially as for plane-bed channel flow of a depth of (H-h), where H=upper flow-surface elevation above roughness trough (or duct height for conduit flow).

For widely spaced *k*-type (interactive-flow) roughness (Fig. 1), experimental results indicate that separation occurs at the lee of a rib with reattachment occurring before the next rib [e.g., Barrantes and Madsen (2000) for triangular ribs normal to the flow], and eddies with a length scale of the order of the roughness height are shed from the roughness and penetrate to the boundary-layer edge. For $\lambda/h\geq 8$, the direct numerical simulation (DNS) results of Leonardi et al. (2003) indicate the ribs to be essentially isolated, with recirculation zones immediately upstream and downstream. For *k*-type roughness of $\lambda/h\geq 8$, these simulations indicate reattachment approximately $5h$ downstream of the trailing rib edge, being independent of the trailing rib for $\lambda/h>8$. A

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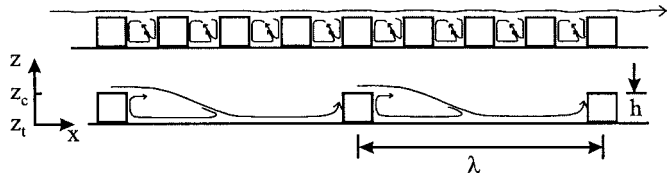

Fig. 1. Roughness geometry and schematic flow paths

transition intermediate roughness condition is often defined as occurring between quasi-smooth and isolated roughness states.

Studies of flows over ribs have primarily been of an industrial focus, namely concerned with flow resistance and heat transfer. Yaglom (1979) in a review of the field to that time found almost no data on velocity profiles, and Jiménez (2004) in a similar review observed a lack of focus on flow structure. As reflected in recent air-flow studies [including Bandyopadhyay (1987); Bandyopadhyay and Watson (1988); and Huckle et al. (1999)], water-flow studies [including Djenidi et al. (1999); Barrantes and Madsen (2000); and Pokrajac et al. (2003)], and numerical simulations [including Liou et al. (1993); Yang and Ferziger (1993); Cui et al. (2000); Ikeda and Durbin (2002); Leonardi et al. (2003); and Stoesser and Rodi (2004)] most researchers have studied only a limited number of rib roughness configurations, and data were typically taken at limited positions within the rib cycle (Cui et al. 2000).

Recognizing the lack of focus to date on flow details over two-dimensional repeated-rib roughness, a series of experiments was thereby undertaken to assess the variation with rib spacing ($\lambda/h=1-16$) of overlying flow structure and dynamics, with a principal focus on velocity and stress distributions and the momentum balance. The following section details averaging measures central to understanding flows over heterogeneous bed roughness. Parameterization of the rough-wall velocity profile is then outlined, the experiments undertaken are described, and the results and implications of the study are presented.

Double-Averaging Methodology (DAM) Framework

For steady water flow over fixed boundaries, the double- (time- and space-) averaged continuity and momentum conservation equations reduce to (Raupach et al. 1991; Finnigan 2000; Nikora 2004; Nikora et al. 2007a,b)

$$\frac{\partial \phi\langle \bar{u}_i\rangle}{\partial x_i}=0 \tag{1}$$

$$\langle \bar{u}_j\rangle\frac{\partial\langle \bar{u}_i\rangle}{\partial x_j}=g_i-\frac{1}{\phi\rho}\frac{\partial\phi\langle\bar{p}\rangle}{\partial x_i}-\frac{1}{\phi}\frac{\partial\phi\langle\overline{u_i'u_j'}\rangle}{\partial x_j}-\frac{1}{\phi}\frac{\partial\phi\langle\tilde{u}_i\tilde{u}_j\rangle}{\partial x_j}+\frac{1}{V_f\rho}\int\int_{S_{\text{int}}}\bar{p}n_i dS-\frac{1}{V_f}\int\int_{S_{\text{int}}}\left(\nu\frac{\partial\bar{u}_i}{\partial x_j}\right)n_j dS \tag{2}$$

where the tensor notation is used with the Einstein convention, which prescribes a summation over each repeated index. Here the spatial averaging is carried out over a thin slab parallel to the mean bed having small thickness dz. The planar extent of the thin-slab averaging volume V_o is typically chosen to be much larger than roughness or flow geometry scales, but smaller than larger geometric features, such as channel curvature and widening or narrowing. When these averages do not change along the flow, the flow can be considered uniform. In the above equations, instantaneous velocity is decomposed as $u_i=\langle\bar{u}_i\rangle+\tilde{u}_i+u_i'$; the straight overbar and angle brackets denote the time and spatial averaging of flow variables, respectively; the wavy overbar denotes spatial fluctuations in the time-averaged flow variable; the prime denotes temporal fluctuations; $\phi=V_f/V_o$=roughness geometry function ($1\geq\phi\geq 0$); V_f=volume of fluid within V_o; S_{int}=roughness–fluid surface interface within the averaging volume; u_i=ith velocity component (in direction x_i); g=gravity; ρ=fluid density; p=pressure; n_i=ith component of the unit vector normal to the surface element dS and directed into the fluid; ν=fluid kinematic viscosity; $\rho\langle\overline{u_i'u_j'}\rangle$=spatially averaged Reynolds stress; and $\rho\langle\tilde{u}_i\tilde{u}_j\rangle$=form-induced (dispersive) stress. The right-handed coordinate system is implied herein, i.e., the x axis (u velocity component) is oriented along the main flow parallel to the averaged bed, the y axis (v velocity component) is oriented to the left bank, and the z axis (w velocity component) is pointing toward the water surface. In Eq. (2), viscous stresses have been ignored (turbulent rough-bed flow) except at the roughness–fluid interface where they contribute to the skin friction.

Integrating the double-averaged momentum conservation equation in the principal flow direction ($i=1$) from any level z to the water surface (z_{ws}), equivalent to simply summing the thin horizontal slabs, gives

$$gS_b\int_z^{z_{\text{ws}}}\phi dz=\int_z^{z_{\text{ws}}}\left[\phi\langle\bar{u}_i\rangle\frac{\partial\langle\bar{u}\rangle}{\partial x_i}+\frac{1}{\rho}\frac{\partial\phi\langle\bar{p}\rangle}{\partial x}+\frac{\partial\phi\langle\overline{u'u_i'}\rangle}{\partial x_i}+\frac{\partial\phi\langle\tilde{u}\tilde{u}_i\rangle}{\partial x_i}\right]dz-\int_z^{z_c}\frac{1}{V_o}\int\int_{S_{\text{int}}}\left[\frac{\bar{p}}{\rho}n_{i=1}-\nu\frac{\partial\bar{u}}{\partial x_i}n_i\right]dSdz \tag{3}$$

where S_b=mean bed slope; $g_{i=1}=gS_b$ is used; and z_c=rib crest elevation. For uniform flow ($\partial/\partial x=0$) with no secondary currents ($\bar{v}=\bar{w}=\partial/\partial y=0$) this becomes

$$\rho gS_b\int_z^{z_{\text{ws}}}\phi dz=-\rho\phi\langle\overline{u'w'}\rangle(z)-\rho\phi\langle\tilde{u}\tilde{w}\rangle(z)-\int_z^{z_c}\frac{1}{V_o}\int\int_{S_{\text{int}}}\left[\bar{p}n_{i=1}-\mu\frac{\partial\bar{u}}{\partial x_i}n_i\right]dSdz \tag{4}$$

where μ=dynamic fluid viscosity; $\nu=\mu/\rho$; and $\langle\overline{u'w'}\rangle(z_{\text{ws}})=\langle\tilde{u}\tilde{w}\rangle(z_{\text{ws}})=0$. This equation can be seen to describe the flux of gravity-induced momentum via fluid stresses (i.e., Reynolds and form-induced stresses) to the boundary where the momentum is removed through form drag and skin friction (for a fixed boundary). For general flows described by Eq. (3), this transfer of momentum also occurs through secondary currents and flow nonuniformity, represented by the additional terms of Eq. (3). In the present experiments, flows were uniformly accelerating, and therefore Eq. (3) can alternatively be expressed as

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$$gS_{\mathrm{ws}}\int_z^{z_{\mathrm{ws}}}\phi dz=\int_z^{z_{\mathrm{ws}}}\left[\phi\langle\bar{u}_i\rangle\frac{\partial\langle\bar{u}\rangle}{\partial x_i}+\frac{\partial\phi\langle\overline{u'u_i'}\rangle}{\partial x_i}+\frac{\partial\phi\langle\tilde{u}\tilde{u}_i\rangle}{\partial x_i}\right]dz$$
$$-\int_z^{z_c}\frac{1}{V_o}\int\int_{S_{\mathrm{int}}}\left[\frac{\bar{p}}{\rho}n_{i=1}-\nu\frac{\partial\bar{u}}{\partial x_i}n_i\right]dSdz \qquad (5)$$

where the gravity and fluid pressure gradient terms have been combined and S_{ws}=slope of the water surface.

Where measurements have only been made over a single roughness wavelength, as was done here, the averaging volume must be limited to this single wavelength, and in the double-averaging approach, all averaged quantities must be assumed not to vary in the flow direction. In order to determine nonuniformity effects for the measured flow field, the measuring window must extend over more than one wavelength, preferably at least two or three wavelengths.

As will be shown below, DAM provides a consistent framework within which flow structure over boundary roughness can be understood and parameterized (e.g., Nikora et al. 2001, 2004) and a consistent picture of flow–boundary interaction can be obtained.

Parameterizing the Velocity Profile

In line with previous boundary-layer characterization, and based on analysis of the double- (time- and space-) averaged equations, Nikora et al. (2001, 2004) propose that rough-wall flow of $\delta/h\gg 1$ may be subdivided into four regions: (1) the interfacial (wall) sublayer (influenced by individual roughness elements) between roughness crests and troughs; (2) a form-induced sublayer just above the roughness crests (where the flow may still be influenced by roughness elements); (3) the logarithmic layer; and (4) the outer layer, where viscous effects and form-induced momentum fluxes are negligible and the double-averaged equations are identical to the time- (ensemble-) averaged equations. Together the form-induced and interfacial sublayers form the roughness layer.

In the overlap region between the inner flow, where viscous and wall–roughness effects dominate, and the outer flow, where these effects are minimal, the distribution of the double-averaged along-channel velocity $\langle\bar{u}\rangle$ in a deep free-surface channel flow over a rough boundary can be described (e.g., Raupach et al. 1991)

$$\frac{\langle\bar{u}\rangle}{u_*}=\frac{1}{\kappa}\ln\left(\frac{z}{\lambda_\nu}\right)+f\left(\frac{h}{\lambda_\nu},\frac{L_i}{\lambda_\nu}\right) \qquad (6)$$

where viscous length scale $\lambda_\nu=\nu/u_*$; shear velocity $u_*=(\tau/\rho)^{0.5}$; τ=bed shear stress; von Karman's constant $\kappa=0.4-0.42$; and L_i=roughness plan lengths (e.g., crest length, crest width, and crest spacing). This logarithmic velocity distribution strictly applies only in a thin layer near the bed (e.g., Townsend 1976; Raupach et al. 1991), being found to extend from a lower bound of $\sim$1–5 roughness heights (h) above roughness crests (e.g., Raupach et al. 1991; Jiménez 2004) to an upper limit of $0.1-0.2H$ (Clauser 1956; Yaglom 1979; Perry et al. 1969; Raupach et al. 1991; Jiménez 2004).

The logarithmic velocity profile of Eq. (6) may also depend on roughness submergence $h/\bar{H}$; Jiménez (2004) noted that $h/\bar{H}<0.0125-0.025$ is required before similarity laws can be expected for rough-wall turbulent-flow profiles, flows for greater $h/\bar{H}$ representing essentially flow over obstacles. Despite such theoretical considerations requiring a deep flow for existence of a genuine overlap logarithmic layer (e.g., Raupach et al. 1991), velocities are empirically found to reasonably approximate a logarithmic behavior throughout most of the depth of open-channel flows [e.g., Barrantes and Madsen (2000) for triangular ribs normal to the flow]. The relation

$$\frac{\langle\bar{u}\rangle}{u_*}=\frac{1}{\kappa}\ln\left(\frac{z-d}{\lambda_\nu}\right)+f\left(\frac{h}{\lambda_\nu},\frac{\lambda_\nu}{\bar{H}},\frac{L_i}{\lambda_\nu}\right)=\frac{1}{\kappa}\ln\left(\frac{z-d}{h}\right)+f\left(\frac{h}{\lambda_\nu},\frac{h}{\bar{H}},\frac{L_i}{h}\right) \qquad (7)$$

is consequently adopted for the purposes of the present work (including facilitating comparison with previous studies), where the origin of the z axis is taken to be the cavity base for the present rib roughness and d=elevation of the zero-velocity plane for the logarithmic profile (e.g., Yaglom 1979; Nikora et al. 2002). Evaluation of a more general form of Eq. (7) to incorporate roughness submergence explicitly is the subject of separate ongoing studies.

The departure of Eq. (7) from the profile for a smooth-wall flow is a function of (h/λ_ν, $h/\bar{H}$, L_i/h), namely, (1) the roughness scale relative to the viscous length scale, (2) the relative submergence for the roughness, and (3) the normalized spatial density of the roughness. Increasing hydrodynamic roughness, and deviation from the profile for a smooth wall, can be expected for increasing relative submergence $h/\bar{H}$. Hydrodynamic roughness can also be expected to increase and then decrease with increasing spatial density L_i/h (e.g., Yaglom 1979; Davies 1980).

The relative importance of the factors (h/λ_ν, $h/\bar{H}$, L_i/h) in forming the rough-boundary velocity profile can be expected to change depending on particular combinations of λ_ν, $\bar{H}$, h, and L_i, with λ_ν potentially becoming of less significance for high-Reynolds-number flows of large U (depth-averaged velocity) and $\bar{H}$ (and vice versa), and h potentially becoming less significant for large $\bar{H}$ (large roughness submergence) or close roughness spacing (small L_i). It is in this latter regard that for two-dimensional rib roughness, Perry et al. (1969) argue that for small L_i/h ("d-type" roughness) the principal parameter is $\bar{H}/\lambda_\nu$ in lieu of the central parameter h/λ_ν for "k-type" roughness of larger L_i/h. Contrary to this distinction, Yaglom (1979) provides evidence that h/λ_ν is the appropriate principal parameter for all roughnesses [Raupach et al. (1991) and Djenidi et al. (1999) also express reservations about an alternate scaling for closely-spaced roughness]. For transverse rectangular two-dimensional rib roughness, as studied herein, Raupach et al. (1991) indicate that the relevant plan lengths L_i are crest-to-crest spacing λ and rib length λ_r (or alternatively, cavity length) in the principal x direction.

Experiments

Experiments were carried out in a nonrecirculating glass-sided flume measuring 0.45 m$\times$0.455 m (wide)$\times$20 m. Each test involved centerline measurements of three-dimensional (3D) velocity vectors, water surface profiles (using upward-looking ultrasonic transducers), and bed pressures. Velocity measurements in planes parallel to the centerline were also undertaken in order to assess secondary current effects. The flume bed was horizontal for the testing, measurements of water surface profile (at intervals of 2.5 mm over the length of the test section) pro-

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Table 1. Test Hydraulic Parameters

λ/h	$\bar{H}$ (m)	U (m/s)	F	m (m)	R ($\times 10^3$)	S_{ws} ($\times 10^{-3}$)	u_* (m/s)	f_T
1	0.150	0.285	0.23	0.09	103	0.24	0.0147	0.021
2	0.158	0.271	0.22	0.09	101	0.49	0.0211	0.049
3	0.160	0.267	0.21	0.09	100	0.65	0.0244	0.067
4	0.161	0.265	0.21	0.09	100	0.80	0.0273	0.085
5	0.162	0.264	0.21	0.09	100	0.97	0.0300	0.104
6	0.163	0.263	0.21	0.09	100	1.14	0.0326	0.123
7	0.163	0.262	0.21	0.09	100	1.16	0.0328	0.125
8	0.163	0.262	0.21	0.10	100	1.18	0.0331	0.128
10	0.164	0.261	0.21	0.10	99	1.18	0.0331	0.128
12	0.164	0.261	0.21	0.10	99	1.08	0.0317	0.118
16	0.164	0.260	0.21	0.10	99	1.03	0.0311	0.114

viding hydraulic-gradient and total-energy-gradient information. The single turbulent flow used for testing was of flow-rate $Q=0.28\ \mathrm{m^3/s}$, and a flow depth above rib tops of $(H-h)=0.15$ m. Table 1 summarizes hydraulic parameters for the tests, where centerline Froude number $\mathsf{F}=U/(g\bar{H})^{0.5}$, m=hydraulic radius, Reynolds number $\mathsf{R}=4mU/\nu$, friction factor $f_T=8(u_*/U)^2$, and shear velocity was determined from the hydraulic radius and the measured free-surface slope, $u_*=(gmS_{ws})^{0.5}$. All tests were fully rough turbulent flows of $h/\lambda_\nu>70$.

The test section of hardwood ribs was 7.1 m long, with the aluminum focus rib (of pressure and velocity measurements) located 5.9 m from the upstream end of the test section. The side lengths of the square ribs measured 0.015 m, with the ribs spanning the full flume width. The ratio $H/h=11$ defines the studied flow as flow of intermediate submergence, as suggested in Nikora et al. (2004). Flow velocities were measured using a Sontek MicroADV (microacoustic Doppler velocimeter), and bed pressures were assessed using a Validyne differential pressure transducer (8 mm full scale reading) connected to 1 mm diameter tappings on the rib. Velocities were measured at 50 Hz for 90 s at each measurement point, at 2–20 mm vertical spacings (focusing near the rib), and 2–40 mm horizontal spacings (increasing with rib spacing) from the downstream edge of the focus rib to the same edge of the immediately upstream rib. Pressure measurements were made on the vertical focus–rib surfaces at 5, 7.5, and

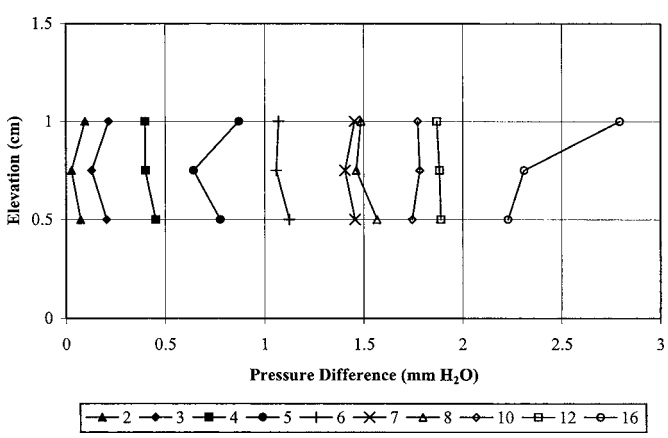

Fig. 2. Pressure differences across vertical rib faces (legend values are λ/h)

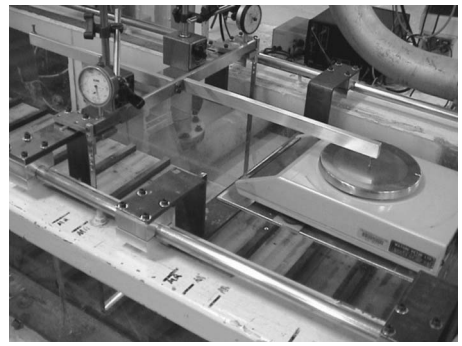

Fig. 3. Direct measurement of rib forces

10 mm from the rib top. With free-surface slopes measured for the tests, assessments of the β parameter of Song and Graf (1994, 1996) showed that nonuniformity effects on flow structure are minor (e.g., a few percent for the present tests). Additional tests at a bed slope of 0.06% were carried out, confirming the relatively minor influence of flow nonuniformity on the measured flow profiles. Rib spacings tested were $\lambda/h=1-16$ (Table 1). These spacings were designed to span d-type and k-type roughness, with a maximum resistance to flow expected within the range of λ/h tested.

For each of the tests, Reynolds and form-induced stresses and secondary current effects were determined from the measured velocities. The skin friction for the ribs was estimated assuming turbulent boundary-layer development over the rib surface. The influence of form drag was assessed from integration of the measured differential-pressure distributions of Fig. 2. Based on the measurements, constant pressure distributions were assumed over the roughness height for each test. Confirming this assumption with their DNS simulations of turbulent flow over square ribs of spacing $\lambda/h=10$, Ikeda and Durbin (2002) found pressures to be nearly constant on each of the upstream and downstream faces of the ribs, except for the region near the upstream upper corner of each rib. The DNS simulations of Leonardi et al. (2003) for $\lambda/h=4-11$ indicate the same patterns of pressure distribution. Leonardi et al. (2003) acknowledge the value of the present pressure measurements, citing such measurements as being difficult to obtain. Subsequent direct measurements of forces on the ribs confirmed the form drag trends and values determined from the measured pressures. These force measurements were carried out (Fig. 3) by suspending the focus rib on a knife edge via thin profile arms, with micrometer adjustment of knife-edge elevation to ensure minimal distance of the rib to the underlying bed while the rib remained free to move. Subsequent testing of slightly varying rib elevation confirmed a minimum influence of this on the measured forces. Balance arms were connected to the knife edge to aid system stability, with one of the balance arms resting, via an extension of adjustable length to ensure minimal load for no flow, on high-precision weight scales. Subtracting the load measurement prior to flow, plus the load for the rib support arms alone in the flow, from the load measured during the flow, the force on the rib was calculated knowing the respective lever arms to the weight scales and the rib via the knife edge (Coleman et al. 2005).

In regard to potential uncertainties in the magnitudes of velocity-based parameters discussed herein, the MicroADV performance characteristics (e.g., www.sontek.com) as used in this study include: (1) an acoustic frequency of 16 MHz; (2) a selected velocity range of ±0.3 m/s; (3) a velocity resolution of

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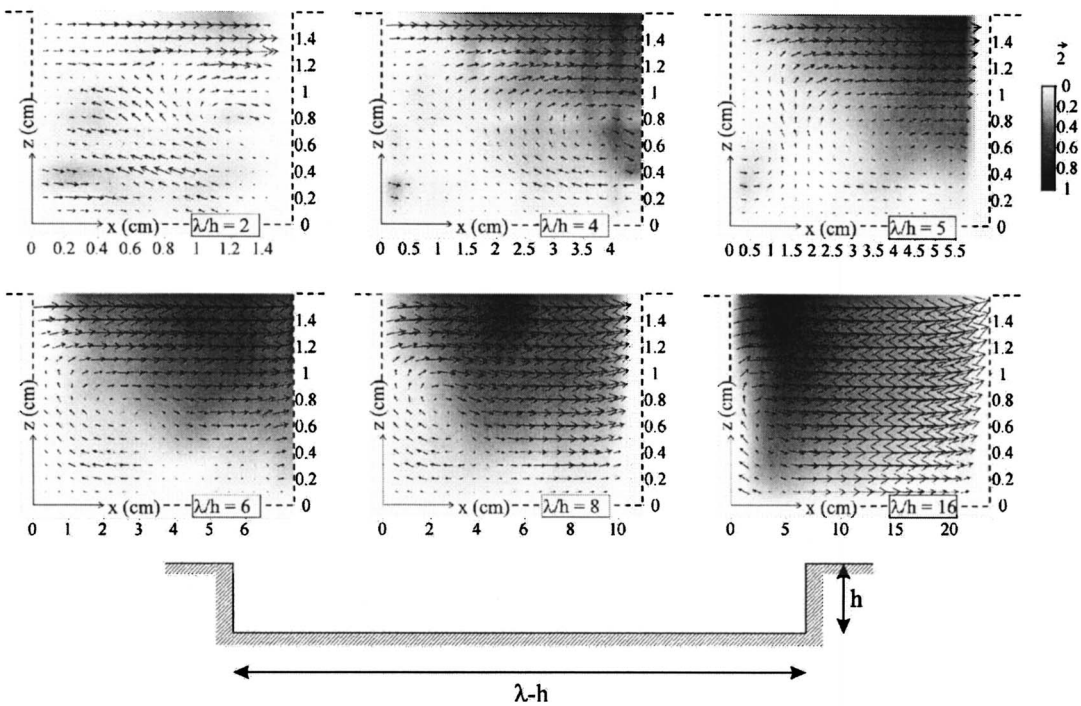

Fig. 4. Velocity vectors $(\bar{u},\bar{w})$ and contours of $\overline{-u'w'}/(g\bar{H}S_{ws})^{0.5}$ [legend values are normalized stress and velocity vector (cm/s) magnitudes]. Each subfigure of distorted axes covers in-cavity region (highlighted in generic sketch, with region boundaries shown for each subfigure) extending $(\lambda-h)$ in x direction by h.

0.1 mm/s; (4) a quoted accuracy of 1% of the measured instantaneous velocity (operating the ADV at 50 Hz); and (5) a standard 4.5 mm tall cylindrical sampling volume of 4.5 mm diameter (located 50 mm away from the sensing elements, enabling measurements to be taken without interfering with the flow). Recognizing that ADV measurements can be significantly affected by probe misalignment, particularly through large-magnitude longitudinal components "leaking into" the smaller lateral and vertical components, efforts were made during the tests to ensure correct probe alignment. Subsequent rotation of measured velocity vectors (e.g., Nikora and Goring 1999) was also used to assess any possible influence of probe misalignment for the present tests. Distances from the bed to the ADV sampling volume were controlled by a screw thread on the ADV mounting and were measured using a pointer gauge referenced to a ruler. The ADV mounting ensured necessary stability in the flow, with no vibration observed visually during experiments. Post-processing was used to remove velocity data of low signal-to-noise ratios and/or correlation values from records, with remaining spikes in the data also eliminated using an acceleration criterion (Goring and Nikora 2002). A typical measurement for a test was determined to be of N=4,500 samples (90 s at 50 Hz), $(\bar{u},\bar{v},\bar{w})=(0.155,0,0)$ ms^{-1}, $(s_u,s_v,s_w)=(0.042,0.030,0.019)$ ms^{-1}, $\overline{u'w'}=-3.8\times10^{-4}$ m^2 s^{-2}, and $\langle\tilde{u}\tilde{w}\rangle=0.2\times10^{-4}$ m^2 s^{-2}, where s_u=standard deviation in u. For this measurement, respective estimated uncertainties in $\bar{u}$, $\tilde{u}$, $\bar{w}$, $\tilde{w}$, $\overline{u'w'}$,and $\langle\tilde{u}\tilde{w}\rangle$ were determined to be $\sigma_{\bar{u}}=\sigma_{\tilde{u}}=6\times10^{-4}$ ms^{-1}, $\sigma_{\bar{w}}=\sigma_{\tilde{w}}=3\times10^{-4}$ ms^{-1}, $\sigma_{\overline{u'w'}}=-15\times10^{-6}$ m^2 s^{-2}, and $\sigma_{\langle\tilde{u}\tilde{w}\rangle}=12\times10^{-6}$ m^2 s^{-2}. Comparable uncertainty magnitudes were determined for other assessed measurements.

Velocity Distribution

The present results (Fig. 4) confirm the circulation patterns described above for the conceptual d-type and k-type roughnesses. Fig. 4 indicates a smooth transition from skimming flow to interactive flow (the conceptual transition from "d-type" to "k-type" roughness) at $\lambda/h=5$ when the above-roughness flow starts to enter the cavity and acts to force the principal clockwise eddy in the cavity to be captured in the rib lee. For d-type roughness, the cavity is found to comprise two principal vortices of opposite rotation, rather than a single principal vortex of clockwise rotation that might be expected for lid-driven cavity flow. The DNS simulations of Leonardi et al. (2003) similarly indicate the cavity of d-type roughness to be occupied by a large recirculation region with a significant secondary vortex, of opposite rotation, toward the lower upstream corner of the cavities. The size in Fig. 4 of the anticlockwise vortex in the rib lee for d-type roughness could be overemphasized by limited ADV resolution of separation and flow reversal in zones of high shear associated with separation zones (McLean, personal communication, 2005). For their turbulent flow over two-dimensional triangular ribs normal to the flow ($\lambda/h=6.67$), Barrantes and Madsen (2000) confirm the potential for ADV overprediction (cf. laser Doppler anemometer measurement) of below-crest streamwise velocity in the wake region immediately downstream of a roughness element, indicating a potential bias below crest level in their spatially averaged profiles of less than +10%. This potential influence of ADV operation needs to be borne in mind in assessing all below-crest velocity-based results presented herein.

Plotting fields of $(\tilde{u},\tilde{w})$ highlights "coherent" structures in the mean flow, namely, organized patterns in mean-flow disturbances

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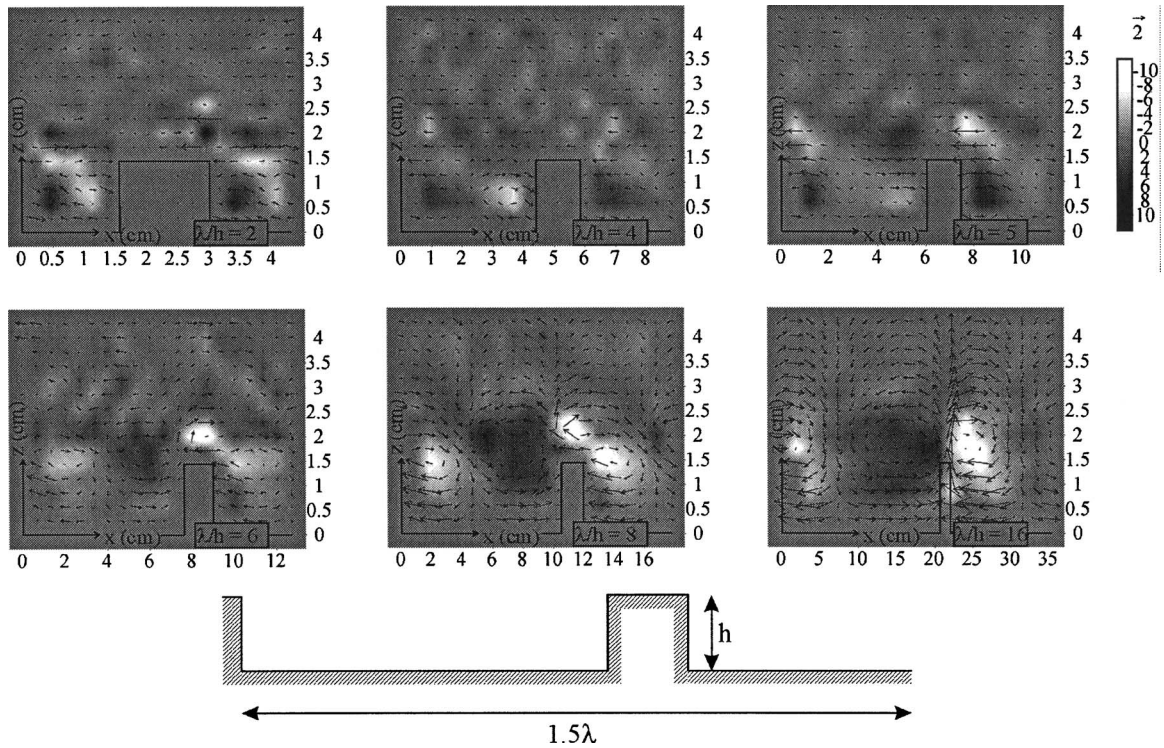

Fig. 5. Fields of $(\tilde{u},\tilde{w})$ and contours of vorticity [legend values are vorticity (s^{-1}) and velocity vector (cm/s) magnitudes]. Each subfigure of distorted axes covers the region (highlighted in generic sketch, with region boundaries shown for each subfigure) extending 1.5λ in x direction by $3h$.

introduced by the ribs. For the present tests (Fig. 5), the flow structure then consists of a pair of principal vortices set up by the ribs, superimposed on an overall double-average profile of interfacial-layer and form-induced-layer sections (discussed below). The principal vortices, of opposite sign, can be seen to swap positions for $\lambda/h<5$ and $\lambda/h>5$ (Fig. 5).

The vertical distribution of double-averaged velocity for flow over ribs displays a logarithmic behavior (Fig. 6) above roughness tops (roughness offset being a function of rib spacing), with a distribution below roughness tops that varies from exponential to linear with increasing rib spacing (Fig. 7). Perry et al. (1969) indicate the change in profile nature for flow over "k-type" roughness to occur at $(1+\alpha)h$ above the roughness trough. For the present tests, the change in profile nature (from logarithmic above crests, to exponential or linear below crests) occurs up to 3 mm ($\alpha=0.2$) above the roughness tops [$0.2>(z-z_c)/h>0$, where z_c=rib crest elevation] for all rib spacings, with no apparent trend with spacing.

Within the interfacial layer between roughness crests and troughs, Nikora et al. (2004) propose that the double-averaged velocity profile can take several forms dependent on flow conditions and roughness geometry: constant; exponential; linear; or a combination of these. Coleman et al. (2006) found a linear profile for water flow over fluvial sand waves of $\lambda/h\approx 20$, and a linear to slightly exponential form for fluvial sand waves of $\lambda/h\approx 10$. Raupach et al. (1991) found an exponential profile for within-canopy air flows of closer roughness spacing. The present results

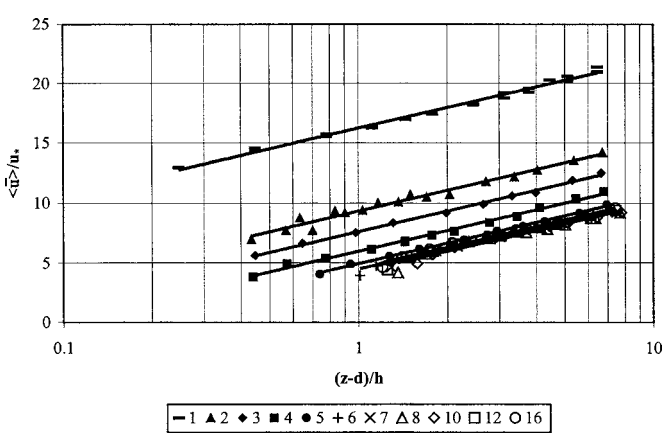

Fig. 6. Logarithmic representation of above-rib velocity profiles (legend values are λ/h)

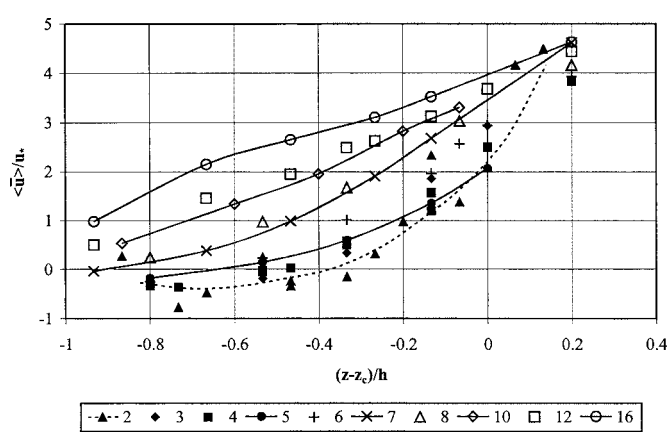

Fig. 7. Velocity profiles below rib tops (legend values are λ/h)

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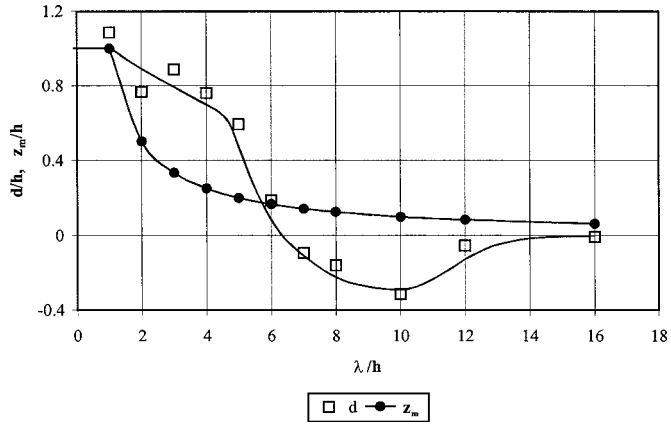

Fig. 8. Logarithmic-profile zero-plane displacement

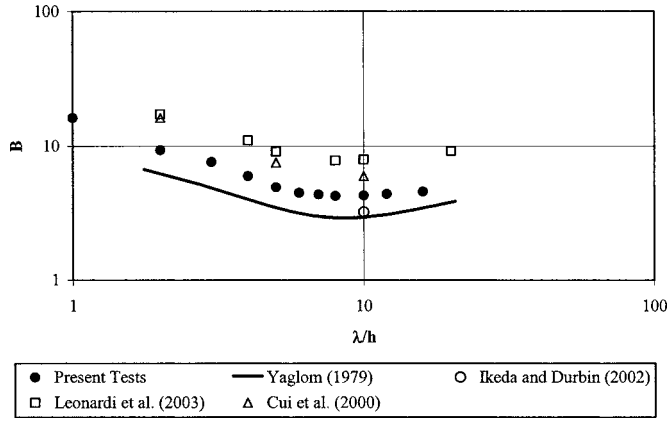

Fig. 9. Logarithmic-profile additive constant

reinforce these trends, an exponential profile occurring (with some flow reversal midrib) for the closely spaced ribs of $\lambda/h < 10$, and a linear profile occurring for the widely spaced ribs of $\lambda/h \geq 10$, with the profile transitioning between the two for increasing λ/h (Fig. 7).

For their fluvial sand-dune profiles, Coleman et al. (2006) found the levels at which the linear velocity profiles go to zero to be only a function of wave steepness h/λ (and not relative submergence $h/\bar{H}$), with $(z_{\langle\bar{u}\rangle=0}/h) = -0.89 \pm 0.03$ and -0.64 ± 0.02 for $\lambda/h = 20$ and 10, respectively ($z=0$ being at the mean bed level). For the present square ribs, this level is confirmed to decrease with increasing λ/h. The level also appears to be a function of roughness shape (e.g., rectangular rib length λ_r in the x direction versus approximately triangular dune profile) however, with this level for the linear profiles being above and below the roughness trough for the sand waves and present ribs, respectively. For the linear profiles of the present tests, $d\langle\bar{u}\rangle/dz$ decreases with increasing λ/h, whereas for the Coleman et al. (2006) linear velocity profiles over dunes, $d\langle\bar{u}\rangle/dz$ was found to increase with increasing λ/h (independent of relative submergence $h/\bar{H}$), again indicating a potential shape influence on the interfacial profile.

For the present ribbed-wall flows of $h/H = 0.09$, a logarithmic behavior is found for the form-induced sublayer above roughness tops (Fig. 6). Correct definition of the effective origin d of the logarithmic profile (e.g., Nikora et al. 2002) is central to correct description and understanding of this layer (e.g., Ikeda and Durbin 2002). Among other fitting methods, Jackson (1981) proposed that the mean value of d should be the centroid of the drag acting on roughness elements (namely, the moment acting on roughness elements divided by the horizontal force on the elements), with $0 < d/h < 1$. Ikeda and Durbin (2002) note, however, the result of $d/h = -0.4$ determined by profile fitting for square-rib-roughened ($\lambda/h = 10$) channel-flow measurements (Hanjalić and Launder 1972). For their DNS simulations of turbulent flow over square ribs of spacing $\lambda/h = 10$, Ikeda and Durbin (2002) also found $d/h < 0$, at odds with $d/h = 0.48$ (at the drag centroid) given by the method of Jackson (1981). For the present velocity profiles of u_* determined from measured water-surface profiles, d was adjusted to give $\kappa = 0.4$ in

$$\frac{\langle\bar{u}\rangle}{u_*} = \frac{1}{\kappa}\ln\left(\frac{z-d}{h}\right) + B \tag{8}$$

The variations in logarithmic-profile origin d and mean bed elevation z_m determined for the present tests are shown in Fig. 8. The origin is effectively at the rib top for no cavity ($\lambda/h = 1$). The origin then decreases with increasing rib spacing to a minimum of $d/h = -0.31$ at $\lambda/h = 10$ and then increases to be effectively at the base of the cavity as rib spacing increases further to give large cavities with isolated roughness. The present results seem physically appropriate, and are consistent with $d/h = -0.4$ determined by Hanjalić and Launder (1972) for square-rib-roughened channel flow of $\lambda/h = 10$, and with $d/h = -0.1$ to -0.5 as determined by Ikeda and Durbin (2002) for modeling of transverse square ribs of $\lambda/h = 10$ and $h/H = 0.059$. The present values of d differ from the equivalent predictions of Jackson (1981) presented by Leonardi et al. (2003) for $h/H = 0.1$ and $\lambda/h = 1.33-20$ of $d/h = 0.43$ to 1, with d/h decreasing with increasing λ/h.

The values of the additive constant B determined for the present tests ($h/H = 0.09$) are plotted in Fig. 9 along with respective smooth-upper-wall numerical-model results of Ikeda and Durbin (2002) for $h/H = 0.06$ [and $d=0$, Ikeda and Durbin (2002) noting the additive constant to increase with decreasing d], and Cui et al. (2000) and Leonardi et al. (2003) for $h/H = 0.1$. Also shown is the Kader (1977) predictive expression for $h/H = 0$ [solid line, as given in Yaglom (1979)], namely, $B = 8.5e^{(-0.4\sigma)} + \sigma^{0.45}$, where the geometric parameter $\sigma = (\lambda/h)^{1/3}\ (\lambda/h - 1)^{2/3}$ for the present two-dimensional transverse square-rib roughness. As can be seen, the present results are consistent with literature indications of magnitudes and trends for B, with B achieving a minimum for $\lambda/h \approx 8$, and with B expected to increase for increasing flow blockage arising with both increasing h/H and also imposition of an upper flow boundary resisting flow.

The variation in B of Fig. 9 reflects the relative shift of the logarithmic profile (Fig. 6) from that for a smooth wall; Figs. 6 and 8 then confirm the indication of f_T values in Table 1 that maximum drag due to wall roughness occurs at $\lambda/h \approx 8$. Measured wall-averaged drag indicates similar trends in flow resistance, with maximum flow resistance confirmed to occur for $\lambda/h = 7-8$ (Fig. 10, where b=channel width). This drag was determined from the earlier-described direct measurements of forces on the ribs. In addition, form drag per unit bed area (Fig. 10) was determined by integrating the measured differential-pressure distributions of Fig. 2. With the surface friction force on a bar being relatively insensitive to bar spacing, this force per unit bed area (the difference between the total and form drag curves on Fig. 10) can be seen to appropriately decrease with increasing bar spacing, becoming relatively negligible for k-type roughness [as noted by Cui et al. (2000); Ikeda and Durbin (2002)]. Early studies of flow

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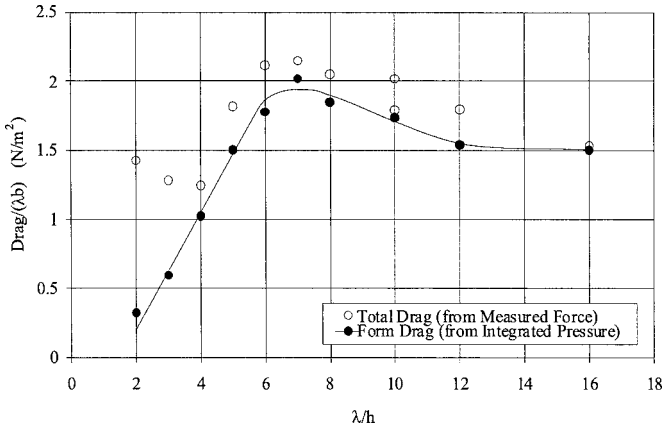

Fig. 10. Wall-averaged-drag variation with rib spacing

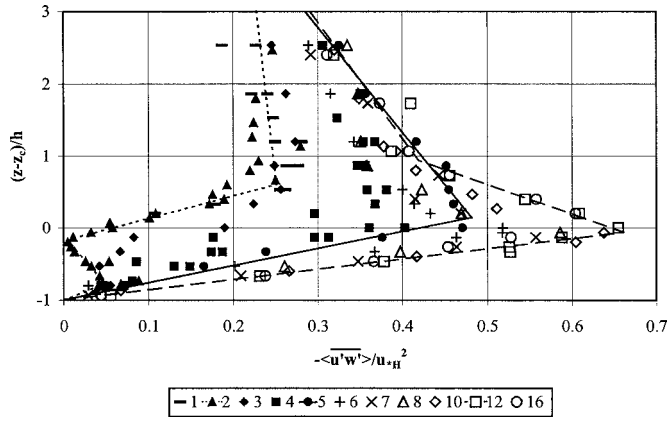

Fig. 11. Normalized Reynolds stress (legend values are λ/h)

resistance for two-dimensional square-rib roughness indicate maximum drag on the flow to occur for $\lambda/h \approx 5$ (Bettermann 1966; Liu et al. 1966) to $\lambda/h \approx 10$ (Davies 1980). These earlier data are fairly coarse, however, and permit the finding herein of $\lambda/h \approx 8$. Through numerical modeling, Leonardi et al. (2003) similarly find form drag per unit area to be a maximum for $\lambda/h=8$ (independent of h/H and Reynolds number), decreasing for greater (reduced flow obstruction) and lesser (more roughness sheltering) λ/h, respectively. In his review, Jiménez (2004) likewise concludes that the maximum resistance to flow occurs for rib steepness $\lambda/h \approx 6.67$.

Further work, with widely varying h/H, λ_r/h, and λ/h, is required to clarify both the theoretical bases of the velocity distributions in the interfacial and form-induced layers, and also detailed "universal" relations that can be adopted.

Momentum Flux

Fields of between-rib (or within-cavity) normalized Reynolds stresses determined for the present tests are shown in Fig. 4. Peak Reynolds stress occurs at the upstream edge of the rib top for $\lambda/h=2$, remaining there with increased Reynolds stresses extending into the cavity zone as λ/h increases to $\lambda/h=5$. For larger rib spacing (larger λ/h), the peak in Reynolds stress remains $\sim 3.3h$ downstream of the trailing rib edge at the level of the rib top. This is consistent with the peak in Reynolds stress over a solid sinusoidal boundary of $\lambda/h=10$ being observed (Hudson et al. 1996) to be midcavity and at the center of the shear layer that separates from the upstream wave crest. In their numerical modeling, Cui et al. (2000) similarly found peak Reynolds stress values to occur in the shear layer over the separation zone downstream from the rib. The results for larger rib spacing are also consistent with the results of Coleman et al. (2006) for flow over fluvial sand dunes, with peak Reynolds stresses occurring in the shear layer associated with the separation zone, consistent with shear-layer instabilities and vortex shedding off this layer.

Variations of spatially averaged Reynolds stress with elevation are given in Fig. 11 for the present tests. In keeping with the form of Eq. (5), stresses and momentum flux components are normalized herein by $\rho u_{*H}^2 = \rho g \bar{H} S_{ws}$. For $\lambda/h \leq 5$, Reynolds stress above the rib varies approximately linearly with depth and increases with rib spacing. For $\lambda/h \geq 5$, a surge in Reynolds stress (larger for increased rib spacing) is apparent as the rib top is approached from above. In line with flow-resistance indications from the velocity profiles, and conceivable from the results of Fig. 11, this surge and the linear profile would be expected to reach a maximum for $\lambda/h \approx 8$, decreasing thereafter with increasing rib spacing. For $\lambda/h=2$, the Reynolds stress approaches zero near the rib top and then increases toward the boundary due to cavity circulation. With greater rib spacing, the stress distribution transitions to a linear decay (for $\lambda/h>5$) from the rib top to the cavity base. For their modeling of transverse k-type square ribs of spacing $\lambda/h=10$, similar results, including the surge in Reynolds stress in the vicinity of the rib top, were found by Ikeda and Durbin (2002).

Fig. 12 presents variations of normalized spatially averaged form-induced stress with elevation. For $\lambda/h \leq 5$, form-induced stresses are essentially zero above the rib, and increase to a positive peak below the rib, this peak decreasing for increasing rib spacing. For $\lambda/h \geq 6$, form-induced stresses are essentially zero for $(z-z_c)/h \geq 3$, below which a negative peak is found just above the rib top, with a positive peak arising about half a rib height below the rib top, the magnitudes of these peaks increasing with rib spacing. From Figs. 11 and 12, it can be seen that form-induced stress acts to offset the surge in Reynolds stress above the rib tops, and also supplements declining Reynolds stress immediately below the rib top. Where deviations in Reynolds stress from a linear variation above roughness crests are large, knowledge of form-induced stress is thereby indicated to be important in describing or modeling flow over the roughness.

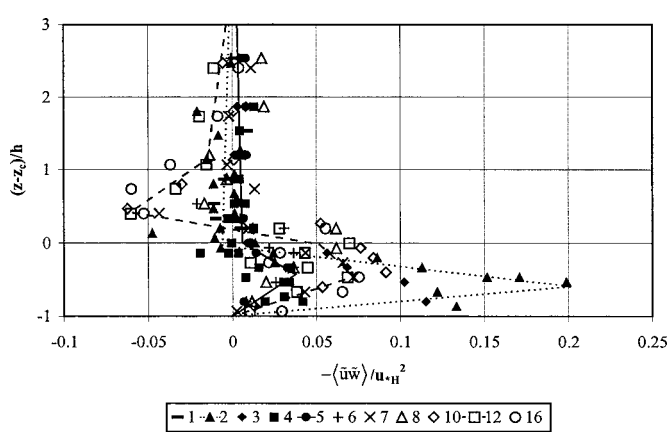

Fig. 12. Normalized form-induced stress (legend values are λ/h)

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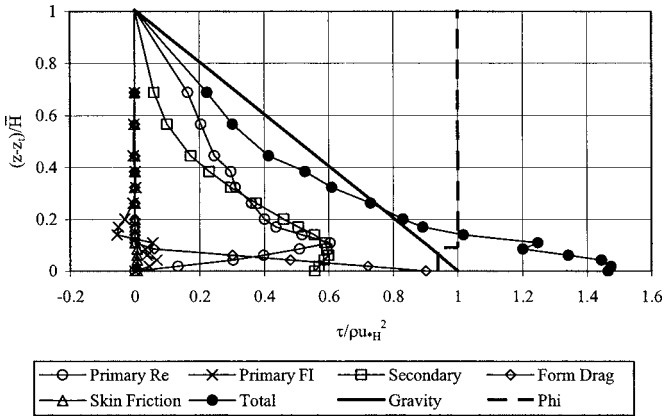

Fig. 13. Momentum flux components and balance for $\lambda/h=16$

Figs. 13 and 14 summarize the variations with elevation of the momentum flux components and momentum balance of Eqs. (4) and (5) for the two tests of maximum and minimum rib spacings λ/h covered in the present work. Primary Reynolds $-\rho\phi\langle\overline{u'w'}\rangle\times(z)$ and form-induced stresses $-\rho\phi\langle\tilde{u}\tilde{w}\rangle(z)$ are presented, along with form drag $-\int_z^{z_c}1/V_o\int\int_{S_{\text{int}}}[\bar{p}n_{i=1}]dSdz$ and skin friction $+\int_z^{z_c}1/V_o\int\int_{S_{\text{int}}}[\mu\,\partial\bar{u}/\partial z]dSdz$ effects, and secondary current terms are collected together (namely, the integral from z to z_{ws} of $[\rho\phi\langle\bar{v}\rangle\partial\langle\bar{u}\rangle/\partial y+\rho\phi\langle\bar{w}\rangle\partial\langle\bar{u}\rangle/\partial z+\rho\,\partial\phi\langle\overline{u'v'}\rangle/\partial y+\rho\,\partial\phi\langle\tilde{u}\tilde{v}\rangle/\partial y]$). For the limited measuring window adopted (refer above), nonuniform flow terms in Eqs. (3) and (5) could not be determined in detail from the measurements for these tests. Also shown for each selected test is the roughness geometry function $\phi(z)$, and the total momentum efflux through flow and boundary effects is compared with influx through gravity $\rho g S_{\text{ws}}\int_z^{z_{\text{ws}}}\phi dz$. Elevations (normalized by the mean flow depth) are referenced to the cavity base z_t. These figures illustrate the use of the double-averaging framework to include secondary current effects on momentum conservation; Eqs. (3)–(5) indicate this process to be similar as, for instance, in Nezu and Nakagawa (1993) for the case of time-averaged flow.

The momentum-flux-component sums of Figs. 13 and 14 are of a concave-upward shape consistent with the accelerating flows of the tests. The gravity-induced flux, equivalent to each sum, is predicted by Eqs. (3)–(5) to be a straight line from zero at the mean water surface to the roughness crest; this line then changes the slope to give unity at the roughness trough

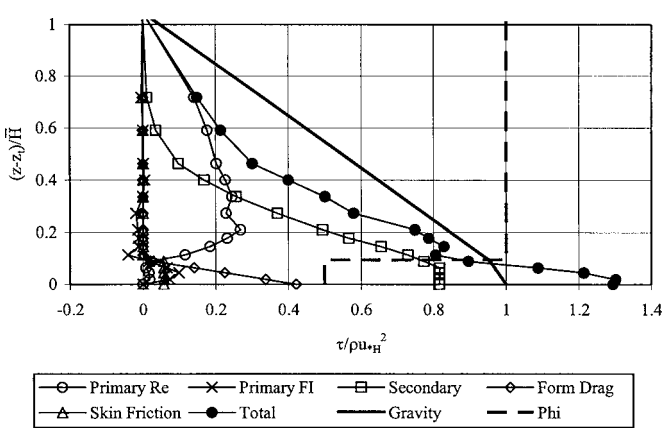

Fig. 14. Momentum flux components and balance for $\lambda/h=2$

(Nikora et al. 2001; 2007a). Within limitations of experimental methods (e.g., ADV sampling-volume and probe-alignment uncertainties, rib alignment, and spacing uncertainties) and measurement accuracies (e.g., ADV accuracies and measurement of very small pressures), the sums of momentum-flux components approximate the theoretical gravity-induced expectations, particularly allowing for unaccounted-for nonuniform-flow effects. Figs. 13 and 14 highlight that secondary currents and flow nonuniformity act to reduce the momentum flux occurring through Reynolds and form-induced stresses for a flow. The results of these figures confirm the potentially significant nature of secondary current effects on momentum transfer and conservation considerations, with conventional extrapolation of measured Reynolds stresses to the mean bed level giving less than half of the true bed shear stress for each run. Secondary flows for these tests arose principally out of a flow aspect (width-to-depth) ratio of 3.0. Without measurement of the resulting crossflows, or of bed pressures, the influence (Figs. 13 and 14) of the resulting secondary currents, not expected for the high relative bed roughness, would not have been detected.

Conclusions

For closely spaced d-type roughness, stable separated vortices occupy the entire cavity between ribs, with minimal disturbance of the outer flow. For widely spaced k-type roughness, reattachment occurs before the next rib, and eddies penetrating to the boundary-layer edge are shed from the roughness. A smooth transition from skimming (d-type) flow to interactive (k-type) flow occurs at a rib spacing of $\lambda/h=5$ (rib wavelength/height).

The structure of flow over rib roughness consists of a pair of principal vortices of opposite sign set up by the ribs (these vortices swap positions for $\lambda/h<5$ and $\lambda/h>5$), superimposed on an overall double-averaged velocity profile. This profile is (quasi-) logarithmic in the form-induced layer above roughness tops, with the profile in the interfacial layer below roughness tops changing from exponential (for ribs of $\lambda/h<10$) to linear ($\lambda/h\geq 10$) to logarithmic ($\lambda/h\gg 10$) with increasing rib spacing. For all rib spacings tested, the change in double-averaged velocity profile nature occurs within $0.2h$ above the roughness tops, with no apparent trend with spacing. In the interfacial layer of a linear velocity profile, velocity gradient and the level at which the velocity goes to zero are functions of λ/h and roughness shape. The logarithmic profile characteristics of $\kappa=0.4$ and the identified trends and values of the additive constant B are consistent with previous studies. The zero plane for the logarithmic formula generally decreases with increasing spacing to a minimum of $d/h=-0.31$ at $\lambda/h=10$, then increases to be effectively at the base of the cavity as rib spacing increases further to give isolated roughnesses. Maximum drag due to wall roughness occurs for $\lambda/h\approx 8$. Further work, with widely varying h/H, λ_r/h, and λ/h, is required to clarify detailed "universal" relations that can be adopted for turbulent flow over rib roughness.

Reynolds stress above the rib varies approximately linearly with depth, and increases with rib spacing; larger rib spacings also give an increasing surge in Reynolds stress in the vicinity of the rib top. This surge and the linear profile reach a maximum for $\lambda/h\approx 8$, decreasing thereafter with increasing rib spacing. Below the rib top, Reynolds stress regenerates with cavity circulation at $\lambda/h=2$, transitioning to a linear decay (for $\lambda/h>5$) from the rib top to the cavity base as λ/h increases. For closely spaced ribs of $\lambda/h\leq 5$, form-induced stresses are essentially zero above the rib,

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and increase to a positive peak below the rib; this peak decreases for increasing rib spacing. For $\lambda/h \geq 6$, form-induced stresses are essentially zero for $(z-z_c)/h \geq 3$, below which a negative peak is found just above the rib top, with a positive peak arising about half a rib height below the rib top; the magnitudes of these peaks increases with rib spacing. Form-induced stress acts to offset any surge in Reynolds stress above the rib tops, and also supplements declining Reynolds stress immediately below the rib top.

The experimentally determined momentum balance, including effects of secondary currents and flow acceleration, is found to agree well with theoretical expectations. It is shown that the knowledge of the effects of form-induced stresses, secondary currents, and flow nonuniformity may be particularly important for describing and modeling flows over roughness elements.

Acknowledgments

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Notation

The following symbols are used in this paper:

B = logarithmic profile additive constant;
b = channel width;
d = elevation of zero-velocity plane for logarithmic velocity profile, pipe diameter;
dz = vertical extent of averaging volume V_o;
F = centerline Froude number, $\mathsf{F}=U/(g\bar{H})^{0.5}$;
f_T = friction factor, $f_T=8(u_*/U)^2$;
g = gravitational acceleration;
H = upper flow–surface elevation above roughness trough (or duct height for conduit flow);
$\bar{H}$ = mean flow depth;
h = roughness height;
k = roughness height;
L_i = roughness plan lengths (e.g., crest length, crest width, and crest spacing);
m = hydraulic radius, $m=(b\bar{H})/(b+2\bar{H})$;
N = number of measurement samples;
n_i = ith component of unit vector normal to surface element dS and directed into fluid;
p = pressure;
Q = flow rate;
R = Reynolds number, $\mathsf{R}=4mU/\nu$;
S_b = mean bed slope;
S_{int} = roughness–fluid surface interface within V_o;
S_{ws} = water surface slope;
s_u = standard deviation in u;
U = depth-averaged velocity;
u_i = $u_i=\bar{u}_i+u_i'=\langle\bar{u}_i\rangle+\tilde{u}_i+u_i'$ = velocity vector (u, v, w), of time-averaged (comprising double-averaged and spatially fluctuating) and temporally fluctuating components;
u_* = shear velocity, $u_*=(\tau/\rho)^{0.5}=(gmS_{\text{ws}})^{0.5}$, $u_{*H}=(g\bar{H}S_{\text{ws}})^{0.5}$;
V_f = volume of fluid within V_o;
V_o = thin-slab averaging volume;
x_i = spatial coordinate (x, y, z), x being along the main flow parallel to the averaged bed, and z being elevation above the between-rib cavity base;
z_c = rib crest elevation;
z_m = mean bed elevation;
z_t = level of the cavity base;
z_{ws} = water surface elevation;
α = normalized (by h) level above roughness crest at which double-averaged velocity profile changes nature;
δ = boundary-layer thickness;
κ = von Karman's constant, $\kappa=0.4$–0.42;
λ = crest-to-crest spacing;
λ_r = rib length in the x direction;
λ_ν = viscous length scale, $\lambda_\nu=\nu/u_*$;
μ = dynamic fluid viscosity;
ν = fluid kinematic viscosity, $\nu=\mu/\rho$;
ρ = fluid density;
σ = geometric parameter;
σ_u = uncertainty in u;
τ = bed shear stress; and
ϕ = roughness geometry function, $\phi=V_f/V_o$ $(1\geq\phi\geq 0)$.

References

Bandyopadhyay, P. R. (1987). "Rough-wall turbulent boundary layers in the transition regime." *J. Fluid Mech.*, 180, 231–266.

Bandyopadhyay, P. R., and Watson, R. D. (1988). "Structure of rough-wall turbulent boundary layers." *Phys. Fluids*, 31(7), 1877–1883.

Barrantes, A. I., and Madsen, O. S. (2000). "Near-bottom flow and flow resistance for currents obliquely incident to two-dimensional roughness elements." *J. Geophys. Res.*, 105(C11), 26,253–26,264.

Bettermann, D. (1966). "Contribution a l'étude de la convection forceé turbulente le long de plaques rugeuses." *Int. J. Heat Mass Transfer*, 9, 153–164.

Clauser, F. H. (1956). *The turbulent boundary layer*, Advances in Applied Mechanics, Vol. 4, Academic, New York.

Coleman, S. E., Nikora, V. I., McLean, S. R., Clunie, T. M., Schlicke, T., and Melville, B. W. (2006). "Equilibrium hydrodynamics concept for developing dunes." *Phys. Fluids*, 18(10), 105104, 1–12.

Coleman, S. E., Nikora, V. I., McLean, S. R., and Schlicke, E. (2005). "Spatially-averaged turbulent flow over square-rib roughness." *4th IAHR Symp. on River, Coastal and Estuarine Morphodynamics*, Urbana, Ill., October 4–7, Taylor and Francis, London, 211–216.

Cui, J., Patel, V. C., and Lin, C-L. (2000). "Large-eddy simulation of turbulent flow over rough surfaces." *Technical Rep. No. 413*, *IIHR*, The Univ. of Iowa, Iowa City, Iowa.

Davies, T. R. H. (1980). "Bedform spacing and flow resistance." *J. Hydr. Div.*, 106(3), 423–433.

Djenidi, L., Elavarasan, R., and Antonia, R. A. (1999). "The turbulent boundary layer over transverse square cavities." *J. Fluid Mech.*, 395, 271–294.

Finnigan, J. J. (2000). "Turbulence in plant canopies." *Annu. Rev. Fluid Mech.*, 32, 519–571.

Goring, D., and Nikora, V. (2002). "De-spiking acoustic Doppler velocimeter data." *J. Hydraul. Eng.*, 128(1), 117–126.

Hanjalić, K., and Launder, B. E. (1972). "Fully developed asymmetric flow in a plane channel." *J. Fluid Mech.*, 51(2), 301–335.

Huckle, E., Pantelic, D., Hu, K., Jones, S., Travkin, V., and Catton, I. (1999). "LDV measurements of the velocity field within a ribbed in-

J. Eng. Mech. 2007.133:194-204.

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ternal duct flow." *Proc., 33rd ASME National Heat Transfer Conf.*, August 14–17, Albuquerque, N.M.

Hudson, I. D., Dykhno, L., and Hanratty, T. J. (1996). "Turbulence production in flow over a wavy wall." *Exp. Fluids*, 20, 257–265.

Ikeda, T., and Durbin, P. A. (2002). "Direct simulations of a rough-wall channel flow." *Rep. No. TF-81*, Dept. of Mechanical Engineering, Stanford Univ., Stanford, Calif.

Jackson, P. S. (1981). "On the displacement height in the logarithmic velocity profile." *J. Fluid Mech.*, 111, 15–25.

Jiménez, J. (2004). "Turbulent flows over rough walls." *Annu. Rev. Fluid Mech.*, 36, 173–196.

Kader, B. A. (1977). "Hydraulic resistance of surfaces covered by two-dimensional roughness at high Reynolds numbers." *Teor. Osn. Khim. Tekhnol.*, 11, 393–404.

Leonardi, S., Orlandi, P., Smalley, R. J., Djenidi, L., and Antonia, R. A. (2003). "Direct numerical simulations of turbulent channel flow with transverse square bars on one wall." *J. Fluid Mech.*, 491, 229–238.

Liou, T. M., Hwang, J-J., and Chen, S-H. (1993). "Simulation and measurement of enhanced turbulent heat transfer in a channel with periodic ribs on one principal wall." *Int. J. Heat Mass Transfer*, 36, 507–517.

Liu, C. K., Kline, S. J., and Johnstone, J. (1966). "An experimental study of turbulent boundary layers on rough walls." *Rep. No. MD-15*, Dept. of Mechanical Engineering, Stanford Univ., Stanford, Calif.

Nezu, I., and Nakagawa, H. (1993). *Turbulence in open-channel flows*, IAHR Monograph Series, A.A. Balkema, Rotterdam.

Nikora, V. (2004). "Spatial averaging concept for rough-bed open-channel and overland flows." *Proc., Sixth Int. Conf. on Hydro-Science and Engineering*, Brisbane, Australia, National Centre for Computational Hydroscience and Engineering, Univ. of Mississippi, Miss. (CD-Rom).

Nikora, V., and Goring, D. G. (1999). "Effects of bed mobility on turbulence structure." *NIWA Internal Rep. No. 48*, NIWA, Christchurch, New Zealand.

Nikora, V., Goring, D., McEwan, I., and Griffiths, G. (2001). "Spatially-averaged open-channel flow over a rough bed." *J. Hydraul. Eng.*, 127(2), 123–133.

Nikora, V., Koll, K., McEwan, I., McLean, S., and Dittrich, A. (2004). "Velocity distribution in the roughness layer of rough-bed flows." *J. Hydraul. Eng.*, 130(10), 1036–1042.

Nikora, V., Koll, K., McLean, S., Dittrich, A., and Aberle, J. (2002). "Zero-plane displacement for rough-bed open-channel flows." *Proc., Int. Conf. on Fluvial Hydraulics, River Flow 2002*, Louvain-la-Neuve, Belgium, September 4–6, Taylor and Francis, London, 83–92.

Nikora, V. I., McLean, S. R., Coleman, S. E., Pokrajac, D., McEwan, I., Campbell, L., Aberle, J., Clunie, T. M., and Koll, K. (2007a). "Double-averaging concept for rough-bed open-channel and overland flows: Applications."*J. Hydrol. Eng.*, in press.

Nikora, V. I., McEwan, I., McLean, S. R., Coleman, S. E., Pokrajac, D., and Walters, R. (2007b). "Double-averaging concept for rough-bed open-channel and overland flows: Theoretical background."*J. Hydrol. Eng.*, in press.

Perry, A. E., Schofield, W. H., and Joubert, P. N. (1969). "Rough wall turbulent boundary layers." *J. Fluid Mech.*, 37(2), 383–413.

Pokrajac, D., Campbell, L. J., Manes, C., Nikora, V. I., and McEwan, I. K. (2003). "Spatially averaged flow over ribbed roughness: A new application of quadrant analysis." *Proc., 30th IAHR Congress*, Theme C, Thessaloniki, Greece, 103–110.

Raupach, M. R., Antonia, R. A., and Rajagopalan, S. (1991). "Rough-wall turbulent boundary layers." *Appl. Mech. Rev.*, 44(1), 1–25.

Simpson, R. L. (1973). "A generalized correlation of roughness density effects on the turbulent boundary layer." *AIAA J.*, 11, 242–244.

Song, T., and Graf, W. H. (1994). "Nonuniform open-channel flow over a rough bed." *J. Hydrosci. Hydr. Eng.*, 12(1), 1–25.

Song, T., and Graf, W. H. (1996). "Velocity and turbulence distribution in unsteady open-channel flows." *J. Hydraul. Eng.*, 122(3), 141–154.

Stoesser, T., and Rodi, W. (2004). "LES of bar and rod roughened channel flow." *Proc., Sixth Int. Conf. on Hydro-Science and Engineering*, Brisbane, Australia, National Centre for Computational Hydroscience and Engineering, Univ. of Mississippi, Miss. (CD-Rom).

Tani, J. (1987). "Turbulent boundary layer development over rough surfaces." *Perspectives in turbulence studies*, Springer, Berlin.

Townes, H. W., and Sabersky, R. H. (1966). "Experiments on the flow over a rough surface." *Int. J. Heat Mass Transfer*, 9, 729–738.

Townsend, A. A. (1976). *The structure of turbulent shear flow*, 2nd Ed., Cambridge Univ., Cambridge, England.

Yaglom, A. M. (1979). "Similarity laws for constant-pressure and pressure-gradient turbulent wall flows." *Annu. Rev. Fluid Mech.*, 11, 505–540.

Yang, K. S., and Ferziger, J. H. (1993). "Large-eddy simulation of turbulent obstacle flow using a dynamic subgrid-scale model." *AIAA J.*, 31, 1406–1413.

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J. Eng. Mech. 2007.133:194-204.